

Kaluza-Klein Theory

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Introduction

We have the electromagnetic four-potential

$$A = -\phi dt + A_1 dx + A_2 dy + A_3 dz$$

where $\vec{E} = -\nabla\phi$ and $\vec{H} = \nabla \times \vec{A}$. Through this we can define the electromagnetic tensor

$$F_{\mu\nu} = \frac{\partial}{\partial x^\mu} A_\nu - \frac{\partial}{\partial x^\nu} A_\mu$$

so

$$[F_{\mu\nu}] = \begin{bmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & H_3 & -H_2 \\ E_2 & -H_3 & 0 & H_1 \\ E_3 & H_2 & -H_1 & 0 \end{bmatrix}.$$

Setup

Suppose we have a 4D spacetime with a metric g , a four-potential A , and a scalar field ψ . For simplicity we will take $\psi = 1$. We wish to extend this to a 5D spacetime.

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We use $(x^0, x^1, x^2, x^3) = (t, x, y, z)$ for the coordinates of the original 4-manifold and $x^5 = w$ for the fifth coordinate.

Greek letters (such as μ or ν) represent four-dimensional indices $(0, 1, 2, 3)$ and capital letters (such as M or N) represent five-dimensional indices.

The Kaluza-Klein Hypothesis

We extend this to a 5D spacetime with the metric

$$h = (g_{\mu\nu} + \psi^2 A_\mu A_\nu) dx^\mu dx^\nu + \psi^2 A_\mu dx^5 dx^\mu + \psi^2 A_\nu dx^\nu dx^5 + \psi^2 (dx^5)^2.$$

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The dual metric is given by

$$g^{\mu\nu} \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\nu} - A^\mu \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^5} - A^\nu \frac{\partial}{\partial x^5} \frac{\partial}{\partial x^\nu} + \left(A^\sigma A_\sigma + \frac{1}{\psi^2} \right) \left(\frac{\partial}{\partial x^5} \right)^2.$$

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The Ricci Scalar

We can use the hypothesized metric to compute the equations of motion and the Einstein Field equations. After computing the Christoffel symbols and the Ricci curvature tensor we arrive at the “Kaluza-Klein” miracle:

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We can use the hypothesized metric to compute the equations of motion and the Einstein Field equations. After computing the Christoffel symbols and the Ricci curvature tensor we arrive at the “Kaluza-Klein” miracle:

$$\widehat{R} = R - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}.$$

This allows us to express the Einstein field equations for the metric h and derive some consequences.

Einstein Field Equations

From the EFE we get the following:

$$\begin{aligned}R &= \frac{3}{4}F_{\sigma\tau}F^{\sigma\tau} \\ \nabla_\lambda(F^{\lambda\sigma}) &= -\frac{1}{4}A^\sigma F_{\tau\lambda}F^{\tau\lambda} \\ R_{\mu\nu} &= \frac{1}{4}g_{\mu\nu}F_{\sigma\tau}F^{\sigma\tau} + \frac{1}{2}F_\nu^\lambda F_{\lambda\mu} + A_\mu A_\nu F_{\sigma\tau}F^{\sigma\tau} \\ 0 &= F_{\sigma\tau}F^{\sigma\tau} \left(\frac{3}{4} + A^\nu A_\nu \right)\end{aligned}$$

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What is the fifth dimension?

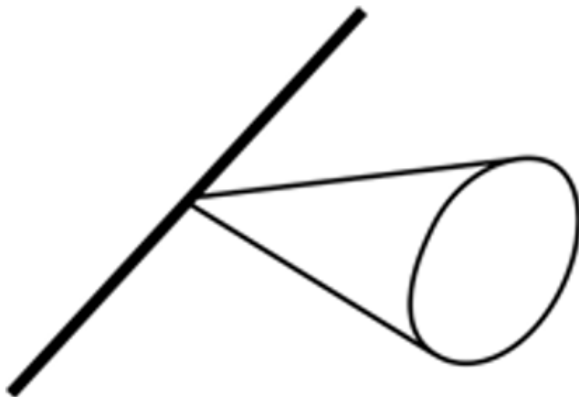
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Klein proposed that the fifth dimension exists and has been unobserved because it has been “compactified” into an extremely small circle.

The compact dimension



References



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